

Project: Simulation of Quantum Decoherence in Multi-Qubit Systems

This project aims to bridge theoretical quantum physics with real-world computational models. This simulation utilizes Density Matrix Formalization, allowing the modeling of non-unitary dynamics in open quantum systems. This application dynamically generates Greenberger-Horne-Zeilinger (GHZ) states for N-qubit systems and applies phase damping to simulate the environmental fragility of the system. The tool visualizes the decay of quantum coherence, tracking the decay of pure quantum entanglement into a classical statistical mixture.

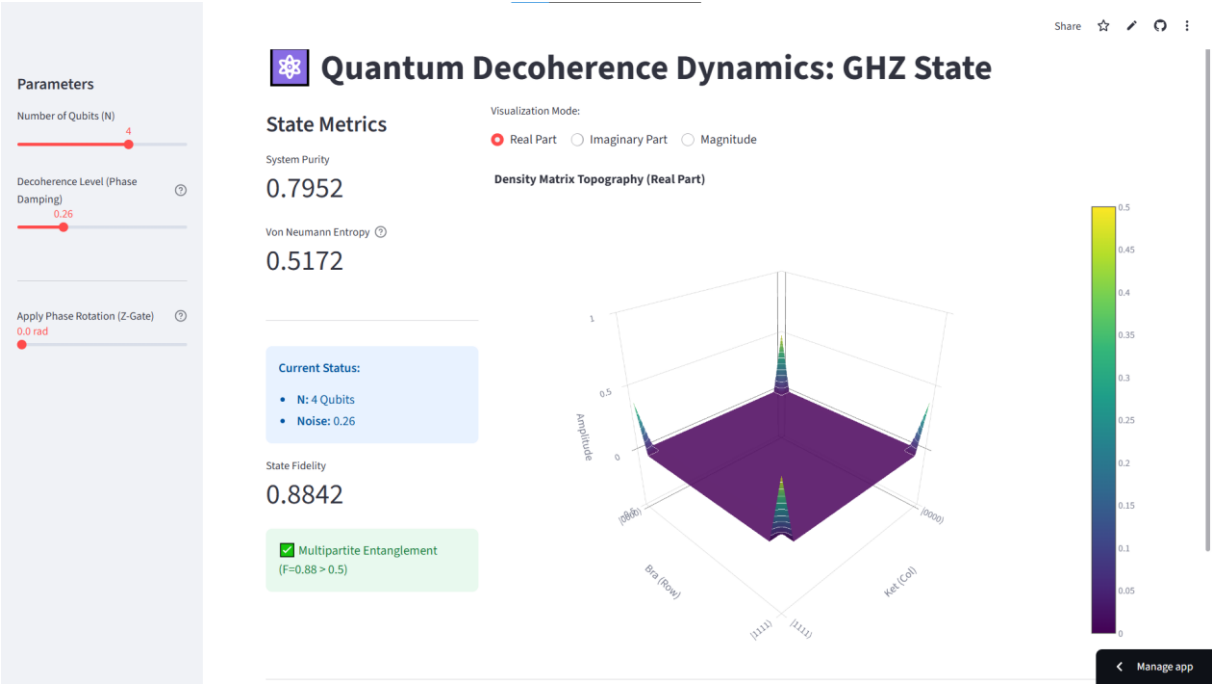


Figure 1: Interface of the quantum entanglement decoherence simulation app. The image visualizes the density matrix of a 4-qubit GHZ state undergoing partial decoherence. The central purple plane indicates the loss of coherence due to phase damping, while the sidebar metrics track the system purity, resulting increase in entropy, and state fidelity.

Technical Interpretation

This simulation utilizes the Qiskit “density_matrix” backend; the system calculates the $2^N \times 2^N$ state matrix. This allows us to derive the system purity and Von Neumann entropy. Deriving a custom scaling algorithm to normalize decoherence probability across variable qubit counts, I ensured consistent noise modeling regardless of the system size(N).

Scalability and Benchmark

In my local execution, the algorithm handled $N \leq 10$ without any error, but the exponential RAM cost of the density matrix (4^N) caused the system to crash at $N = 12$. Due to strict resource limitations of community clouds hosting in Streamlit, this simulation is restricted to 5 qubits.

Visualization and Future Implementations

A 3D interactive surface plot renders the real, imaginary, and magnitude of the density matrix for topological interpretation.

The current version focuses on demonstrating the maximum fragility of GHZ states. I am currently working on simulating both GHZ and W to introduce comparison.

While GHZ states lose all multipartite entanglement upon the loss of a single qubit, the W-states remain entangled and preserve coherence, providing decoherence durability in quantum networks. Visualizing the comparison of both GHZ and W states provides a stronger foundation for understanding the decoherence durability in quantum networks, for both use and advantages.

GitHub Repository for Notes of Self-Studied Courses:

<https://github.com/said-tng/Self-Study-Notes>

GitHub Repository for Code:

https://github.com/said-tng/Quantum_Entanglement_Simulation_On_GHZ_States

Interactive Web Demo Streamlit:

<https://quantum-entanglement-simulation.streamlit.app/>

*QR Code for Web Demo
Streamlit Simulation*

